

Maximum Likelihood Estimation

ECMT 676 (Econometric II) Lecture Notes

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- Let $(\mathbf{x}_i, \mathbf{y}_i)$ denote a random draw from a population, where $\mathbf{x}_i$ and $\mathbf{y}_i$ can both be vectors.
- Suppose we are interested in the distribution of $\mathbf{y}_i$ conditional on $\mathbf{x}_i$, $D(\mathbf{y}_i|\mathbf{x}_i)$.
- If we use a parametric model for a density describing $D(\mathbf{y}_i|\mathbf{x}_i)$, maximum likelihood estimation is natural.
- Some use the label *conditional maximum likelihood estimation (CMLE)* because we are not modeling the distribution of $\mathbf{x}_i$.

- First just start with the case where we have a density for $D(\mathbf{y}_i | \mathbf{x}_i = \mathbf{x})$ for possible outcomes $\mathbf{x}$.
- Contrast with NLS, where we model just the conditional mean. Or LAD, where we model the conditional median. In principle, we can recover all of these features, and more, by modeling the entire distribution.
- Let Θ be the parameter space, a subset of $\mathbb{R}^P$.
- The model of the density is denoted

$$f(\mathbf{y} | \mathbf{x}; \boldsymbol{\theta}), \mathbf{y} \in \mathcal{Y}, \mathbf{x} \in \mathcal{X}, \boldsymbol{\theta} \in \Theta,$$

where we assume this is a proper density for each $\boldsymbol{\theta}$. (That is, it is nonnegative and “integrates” to one.)

- We want to allow $\mathbf{y}_i$ to have any characteristic: continuous, discrete, or possibly both features.

EXAMPLE (Probit): Suppose y_i is a scalar binary response, so it takes just two values, zero and one. Let $\mathbf{x}_i$ be a $1 \times K$ vector with $x_{i1} = 1$ for simplicity.

- Suppose y_i is generated by a linear latent variable model:

$$\begin{aligned}y_i^* &= \mathbf{x}_i \boldsymbol{\theta} + e_i \\ e_i | \mathbf{x}_i &\sim \text{Normal}(0, 1) \\ y_i &= 1 \text{ if } y_i^* > 0 \\ &= 0 \text{ if } y_i^* \leq 0.\end{aligned}$$

If $e_i|\mathbf{x}_i \sim \text{Normal}(0, \sigma^2)$, then $P(y_i = 1|\mathbf{x}_i) = \Phi(\mathbf{x}_i\boldsymbol{\theta}/\sigma)$

We can only identify $\boldsymbol{\theta}/\sigma$.

We can not identify $\boldsymbol{\theta}$ and σ separately

So for identification, we assume $\sigma = 1$.

- A useful shorthand is

$$y_i = 1[y_i^* > 0]$$

where $1[\cdot]$ is the indicator function.

- The data we observe are $(\mathbf{x}_i, y_i)$, and are usually interested in the effects of $\mathbf{x}_i$ on the *response probability*

$$\begin{aligned} P(y_i = 1|\mathbf{x}_i) &= P(y_i^* > 0|\mathbf{x}_i) = P(\mathbf{x}_i\boldsymbol{\theta} + e_i > 0|\mathbf{x}_i) \\ &= P(e_i > -\mathbf{x}_i\boldsymbol{\theta}|\mathbf{x}_i) = 1 - \Phi(-\mathbf{x}_i\boldsymbol{\theta}) = \Phi(\mathbf{x}_i\boldsymbol{\theta}), \end{aligned}$$

where $\Phi(\mathbf{z}) = \int_{-\infty}^{\mathbf{z}} \phi(v)dv$ is the standard normal cdf and $\phi(v) = (2\pi)^{-1/2} \exp(-v^2/2)$ is the standard normal pdf.

- We have now completely characterized the conditional distribution:

$$\begin{aligned}f(1|\mathbf{x}; \boldsymbol{\theta}) &= \Phi(\mathbf{x}\boldsymbol{\theta}) \\f(0|\mathbf{x}; \boldsymbol{\theta}) &= 1 - \Phi(\mathbf{x}\boldsymbol{\theta})\end{aligned}$$

or

$$\begin{aligned}f(y|\mathbf{x}; \boldsymbol{\theta}) &= [1 - \Phi(\mathbf{x}\boldsymbol{\theta})]^{(1-y)}\Phi(\mathbf{x}\boldsymbol{\theta})^y, \quad y = 0, 1 \\&= 0 \text{ if } y \notin \{0, 1\}.\end{aligned}$$

EXAMPLE (Normal) Let $m(\mathbf{x}, \boldsymbol{\theta})$ be the mean function and $v(\mathbf{x}, \boldsymbol{\theta})$ the variance function. Then

$$f(y|\mathbf{x}; \boldsymbol{\theta}) = [2\pi v(\mathbf{x}, \boldsymbol{\theta})]^{-1/2} \exp\{-[y - m(\mathbf{x}, \boldsymbol{\theta})]^2 / [2v(\mathbf{x}, \boldsymbol{\theta})]\}, \quad -\infty < y < \infty$$

- Often the mean and variance parameters are entire separate, but not always.
- As it turns out, the normal distribution identifies the first two moments even if the distribution is not normal. That leads us to the notion of QMLE for the conditional mean and conditional variance jointly. For now assume the density is correctly specified in its entirety.
- If the variance is taken as constant, say $v(\mathbf{x}, \boldsymbol{\theta}) = \sigma^2$, then the Gaussian (normal) MLE of the mean parameters is the NLS estimator. We know NLS is consistent if just the conditional mean is correctly specified. Therefore, we already know that the Gaussian MLE is robust to certain kinds of distributional misspecification.

- The motivation for MLE in introductory statistics is intuitively appealing, but it does not directly lead to a verification of consistency.
- In fact, we will apply the M-estimation results to the objective function

$$q(\mathbf{w}_i, \theta) = -\log f(\mathbf{y}_i | \mathbf{x}_i; \theta)$$

- $\ell_i(\theta) \equiv \log f(\mathbf{y}_i | \mathbf{x}_i; \theta)$ called the *log-likelihood function for observation i* . It is random because it depends on $(\mathbf{x}_i, \mathbf{y}_i)$, but we are interested in it as a function of θ .
- For the theory we should be careful about using θ_o for the “true value”.

- This means $f(\mathbf{y}|\mathbf{x}; \theta_o)$ is the true density of $\mathbf{y}_i$ given $\mathbf{x}_i = \mathbf{x}$ for all $\mathbf{x} \in \mathcal{X}$.
- The (Conditional) Maximum Likelihood Estimator of θ_o , $\hat{\theta}$, solves

$$\max_{\theta \in \Theta} N^{-1} \sum_{i=1}^N \log f(\mathbf{y}_i | \mathbf{x}_i; \theta).$$

- To show consistency of the MLE we need to show that the log likelihood identifies θ_o .
- This follows by the *Kullback-Leibler Information Inequality*.

- The conditional KLII (which can be derived from the conditional Jensen's inequality – See Appendix 13A) states that

$$E[\log f(\mathbf{y}_i|\mathbf{x}_i; \boldsymbol{\theta}_o)|\mathbf{x}_i] \geq E[\log f(\mathbf{y}_i|\mathbf{x}_i; \boldsymbol{\theta})|\mathbf{x}_i], \text{ all } \boldsymbol{\theta} \in \Theta.$$

- Taking the expected value with respect to the distribution of $\mathbf{x}_i$ and using the LIE gives the unconditional KLII:

$$E[\log f(\mathbf{y}_i|\mathbf{x}_i; \boldsymbol{\theta}_o)] \geq E[\log f(\mathbf{y}_i|\mathbf{x}_i; \boldsymbol{\theta})], \text{ all } \boldsymbol{\theta} \in \Theta.$$

- The KLII implies θ_o is one solution.
- For identification, we assume θ_o is the unique solution.
- Uniqueness *depends* on the functional form of the density and the distribution of $\mathbf{x}_i$.
- For example, in the probit and Poisson cases, perfect collinearity in $\mathbf{x}_i$ will violate identification, just as in linear regression.
- Provided $\ell_i(\theta) \equiv \log f(\mathbf{y}_i|\mathbf{x}_i; \theta)$ is continuous in θ and that enough moments of the log likelihood are bounded across θ , the MLE is generally consistent. Apply the M-estimation consistency result directly.

- Discontinuity in the parameter is rare, but happens if, say, the support of the distribution depends on θ , as in the case of a $Uniform(0, \theta)$ distribution. The MLE is usually consistent in such cases, but other arguments are needed.

- A consistency theorem for the M-estimator $(\hat{\theta})$ without compactness and uniform convergence.
- Suppose (i) identification of θ_o ; (ii) $q(w_i, \theta)$ is convex in θ for all w_i ; (iii) $E|q(w_i, \theta)| < \infty$ for all θ .
- Then $\hat{\theta} \xrightarrow{P} \theta_o$.

EXAMPLE (Probit): The log likelihood for a random draw i are

$$\begin{aligned}\ell_i(\boldsymbol{\theta}) &= (1 - y_i) \log[1 - \Phi(\mathbf{x}_i\boldsymbol{\theta})] + y_i \log[\Phi(\mathbf{x}_i\boldsymbol{\theta})] \\ &= (1 - y_i) \log[\Phi(-\mathbf{x}_i\boldsymbol{\theta})] + y_i \log[\Phi(\mathbf{x}_i\boldsymbol{\theta})] \\ q(w_i, \theta) &= -\ell_i(\boldsymbol{\theta})\end{aligned}$$

$q(w_i, \theta)$ is convex since $\log[\Phi(\mathbf{x}_i\boldsymbol{\theta})]$ is concave in θ .

Here we use a property that standard normal CDF is log-concave.

(If the density is log concave, then the CDF is log concave.

$$\log \phi(v) = -(\log 2\pi)/2 - v^2/2)$$

Asymptotic Distribution

- Denote the score of the log likelihood as the $P \times 1$ vector

$$\mathbf{s}_i(\boldsymbol{\theta}) = \mathbf{s}(\mathbf{x}_i, \mathbf{y}_i, \boldsymbol{\theta}) = \nabla_{\boldsymbol{\theta}} \log f(\mathbf{y}_i | \mathbf{x}_i; \boldsymbol{\theta})' = \nabla_{\boldsymbol{\theta}} \ell_i(\boldsymbol{\theta})'$$

Further, the Hessian is still the Jacobian of the score:

$$\mathbf{H}_i(\boldsymbol{\theta}) = \mathbf{H}(\mathbf{x}_i, \mathbf{y}_i, \boldsymbol{\theta}) = \nabla_{\boldsymbol{\theta}} \mathbf{s}_i(\boldsymbol{\theta})$$

- A slight notational change from M-estimation (because we are doing a maximization problem):

$$\mathbf{A}_o = -E[\mathbf{H}_i(\boldsymbol{\theta}_o)]$$

$$\mathbf{A}(\mathbf{x}_i, \boldsymbol{\theta}_o) = -E[\mathbf{H}_i(\boldsymbol{\theta}_o) | \mathbf{x}_i]$$

so that $\mathbf{A}(\mathbf{x}_i, \boldsymbol{\theta}_o)$ is positive semi-definite and $\mathbf{A}_o$ is pd.

- As before, let

$$\mathbf{B}_o = E[\mathbf{s}_i(\boldsymbol{\theta}_o)\mathbf{s}_i(\boldsymbol{\theta}_o)'].$$

- It is useful to verify certain features of the score. First, because $\boldsymbol{\theta}_o$ solves

$$\max_{\boldsymbol{\theta} \in \Theta} E[\log f(\mathbf{y}_i | \mathbf{x}_i; \boldsymbol{\theta}) | \mathbf{x}_i],$$

the score generally satisfies

$$E[\mathbf{s}_i(\boldsymbol{\theta}_o) | \mathbf{x}_i] = \mathbf{0}$$

and so

$$E[\mathbf{s}_i(\boldsymbol{\theta}_o)] = \mathbf{0}.$$

- The condition $E[\mathbf{s}_i(\boldsymbol{\theta}_o)] = \mathbf{0}$ is sometimes referred to as *Fisher consistency*: the MLE solves a population maximization problem. We then use the sample analog.
- Further, under regularity conditions, the *conditional information matrix equality (CIME)* holds. Namely,

$$-E[\mathbf{H}_i(\boldsymbol{\theta}_o)|\mathbf{x}_i] = E[\mathbf{s}_i(\boldsymbol{\theta}_o)\mathbf{s}_i(\boldsymbol{\theta}_o)'|\mathbf{x}_i]$$

which implies the *unconditional information matrix equality (UIME)*:

$$-E[\mathbf{H}_i(\boldsymbol{\theta}_o)] = E[\mathbf{s}_i(\boldsymbol{\theta}_o)\mathbf{s}_i(\boldsymbol{\theta}_o)'].$$

- In the notation of M-estimation,

$$\mathbf{A}_o = \mathbf{B}_o.$$

- Therefore, for correctly specified (conditional) maximum likelihood problems,

$$Avar[\sqrt{N}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_o)] = \mathbf{A}_o^{-1} = \mathbf{B}_o^{-1}.$$

- So, generally, one chooses among three estimators of $Avar(\hat{\boldsymbol{\theta}})$:

$$\left(\sum_{i=1}^N -\mathbf{H}_i(\hat{\boldsymbol{\theta}}) \right)^{-1}, \quad \left(\sum_{i=1}^N \mathbf{A}_i(\hat{\boldsymbol{\theta}}) \right)^{-1}, \quad \left(\sum_{i=1}^N \mathbf{s}_i(\hat{\boldsymbol{\theta}}) \mathbf{s}_i(\hat{\boldsymbol{\theta}})' \right)^{-1}.$$

- The outer product of the score formulation, while computationally simple, can have severe finite-sample bias; usually the standard errors are too small on average.
- The Hessian and expected Hessian forms tend to work well. In leading cases, the expected Hessian form depends only on first derivatives.
- If we entertain the possibility that the conditional density is misspecified, then a sandwich form from M-estimation is required. In econometrics, this notion was popularized by White (1982, *Econometrica*).

- EXAMPLE (Probit): The log likelihood and gradients for a random draw i are

$$\begin{aligned}\ell_i(\boldsymbol{\theta}) &= (1 - y_i) \log[1 - \Phi(\mathbf{x}_i\boldsymbol{\theta})] + y_i \log[\Phi(\mathbf{x}_i\boldsymbol{\theta})] \\ \nabla_{\boldsymbol{\theta}} \ell_i(\boldsymbol{\theta}) &= -(1 - y_i) \mathbf{x}_i \phi(\mathbf{x}_i\boldsymbol{\theta}) / [1 - \Phi(\mathbf{x}_i\boldsymbol{\theta})] \\ &\quad + y_i \mathbf{x}_i \phi(\mathbf{x}_i\boldsymbol{\theta}) / \Phi(\mathbf{x}_i\boldsymbol{\theta}) \\ &= \phi(\mathbf{x}_i\boldsymbol{\theta}) \mathbf{x}_i \frac{-(1 - y_i) \Phi(\mathbf{x}_i\boldsymbol{\theta}) + y_i [1 - \Phi(\mathbf{x}_i\boldsymbol{\theta})]}{\Phi(\mathbf{x}_i\boldsymbol{\theta}) [1 - \Phi(\mathbf{x}_i\boldsymbol{\theta})]} \\ &= \phi(\mathbf{x}_i\boldsymbol{\theta}) \mathbf{x}_i \frac{[y_i - \Phi(\mathbf{x}_i\boldsymbol{\theta})]}{\Phi(\mathbf{x}_i\boldsymbol{\theta}) [1 - \Phi(\mathbf{x}_i\boldsymbol{\theta})]}\end{aligned}$$

- Therefore, the score is

$$\mathbf{s}_i(\boldsymbol{\theta}) = \phi(\mathbf{x}_i; \boldsymbol{\theta}) \mathbf{x}_i' \frac{[y_i - \Phi(\mathbf{x}_i; \boldsymbol{\theta})]}{\Phi(\mathbf{x}_i; \boldsymbol{\theta})[1 - \Phi(\mathbf{x}_i; \boldsymbol{\theta})]}$$

and

$$E[\mathbf{s}_i(\boldsymbol{\theta}_o) | \mathbf{x}_i] = \phi(\mathbf{x}_i; \boldsymbol{\theta}_o) \mathbf{x}_i' \frac{[E(y_i | \mathbf{x}_i) - \Phi(\mathbf{x}_i; \boldsymbol{\theta}_o)]}{\Phi(\mathbf{x}_i; \boldsymbol{\theta}_o)[1 - \Phi(\mathbf{x}_i; \boldsymbol{\theta}_o)]} = \mathbf{0}$$

because $E(y_i | \mathbf{x}_i) = P(y_i = 1 | \mathbf{x}_i) = \Phi(\mathbf{x}_i; \boldsymbol{\theta}_o)$.

- Note that $E[\mathbf{s}_i(\boldsymbol{\theta}_o) | \mathbf{x}_i] = \mathbf{0}$ whenever $E(y_i | \mathbf{x}_i) = \Phi(\mathbf{x}_i; \boldsymbol{\theta}_o)$, even if y_i is not binary. For example, y_i could be a fractional response. (More in Chapter 18 with quasi-MLE.)

- The Hessian has the form

$$\mathbf{H}_i(\theta) = \nabla_{\theta} \mathbf{s}_i(\theta) = \frac{-[\phi(\mathbf{x}_i; \theta)]^2 \mathbf{x}_i' \mathbf{x}_i}{\Phi(\mathbf{x}_i; \theta)[1 - \Phi(\mathbf{x}_i; \theta)]} + \mathbf{L}(\mathbf{x}_i, \theta)[y_i - \Phi(\mathbf{x}_i; \theta)]$$

where $\mathbf{L}(\mathbf{x}_i, \theta)$ is the Jacobian of

$$\frac{\phi(\mathbf{x}_i; \theta) \mathbf{x}_i'}{\Phi(\mathbf{x}_i; \theta)[1 - \Phi(\mathbf{x}_i; \theta)]}.$$

Under correct specification, we can use

$$\mathbf{A}(\mathbf{x}_i, \theta_o) = -E[\mathbf{H}_i(\theta_o) | \mathbf{x}_i] = \frac{[\phi(\mathbf{x}_i; \theta_o)]^2 \mathbf{x}_i' \mathbf{x}_i}{\Phi(\mathbf{x}_i; \theta_o)[1 - \Phi(\mathbf{x}_i; \theta_o)]}$$

- Then,

$$\hat{\mathbf{A}} = N^{-1} \sum_{i=1}^N \frac{[\phi(\mathbf{x}_i; \hat{\boldsymbol{\theta}})]^2 \mathbf{x}_i' \mathbf{x}_i}{\Phi(\mathbf{x}_i; \hat{\boldsymbol{\theta}})[1 - \Phi(\mathbf{x}_i; \hat{\boldsymbol{\theta}})]} \xrightarrow{p} \mathbf{A}_o.$$

So the “usual” asymptotic variance estimator is

$$\left(\sum_{i=1}^N \frac{[\phi(\mathbf{x}_i; \hat{\boldsymbol{\theta}})]^2 \mathbf{x}_i' \mathbf{x}_i}{\Phi(\mathbf{x}_i; \hat{\boldsymbol{\theta}})[1 - \Phi(\mathbf{x}_i; \hat{\boldsymbol{\theta}})]} \right)^{-1},$$

which is easily seen to be positive definite when the inverse exists.

- We can verify the conditional information matrix equality.
- Write $u_i(\theta) = y_i - \Phi(\mathbf{x}_i; \theta)$ and $u_i = y_i - \Phi(\mathbf{x}_i; \theta_o)$. Then

$$\mathbf{s}_i(\theta)\mathbf{s}_i(\theta)' = [\phi(\mathbf{x}_i; \theta)]^2 \mathbf{x}_i' \mathbf{x}_i \frac{[u_i(\theta)]^2}{\{\Phi(\mathbf{x}_i; \theta)[1 - \Phi(\mathbf{x}_i; \theta)]\}^2}$$

so

$$\begin{aligned} E[\mathbf{s}_i(\theta_o)\mathbf{s}_i(\theta_o)' | \mathbf{x}_i] &= [\phi(\mathbf{x}_i; \theta_o)]^2 \mathbf{x}_i' \mathbf{x}_i \frac{E(u_i^2 | \mathbf{x}_i)}{\{\Phi(\mathbf{x}_i; \theta_o)[1 - \Phi(\mathbf{x}_i; \theta_o)]\}^2} \\ &= [\phi(\mathbf{x}_i; \theta_o)]^2 \mathbf{x}_i' \mathbf{x}_i \frac{\text{Var}(y_i | \mathbf{x}_i)}{\{\Phi(\mathbf{x}_i; \theta_o)[1 - \Phi(\mathbf{x}_i; \theta_o)]\}^2} \end{aligned}$$

assuming $E(y_i | \mathbf{x}_i) = \Phi(\mathbf{x}_i; \theta_o)$.

- For a binary response y_i , $\text{Var}(y_i|\mathbf{x}_i) = \Phi(\mathbf{x}_i\boldsymbol{\theta}_o)[1 - \Phi(\mathbf{x}_i\boldsymbol{\theta}_o)]$, and so

$$E[\mathbf{s}_i(\boldsymbol{\theta}_o)\mathbf{s}_i(\boldsymbol{\theta}_o)'|\mathbf{x}_i] = \frac{[\phi(\mathbf{x}_i\boldsymbol{\theta}_o)]^2 \mathbf{x}_i' \mathbf{x}_i}{\Phi(\mathbf{x}_i\boldsymbol{\theta}_o)[1 - \Phi(\mathbf{x}_i\boldsymbol{\theta}_o)]} = -E[\mathbf{H}_i(\boldsymbol{\theta}_o)|\mathbf{x}_i],$$

which is easily seen to be psd for any $\mathbf{x}_i$.

MLE Testing

- We can apply three statistics from M-estimation: Wald, Lagrange Multiplier (score), and Likelihood Ratio.
- Under correct specification of the density we can use the LR statistic:

$$LR = 2(\mathcal{L}_{ur} - \mathcal{L}_r) = 2 \left[\sum_{i=1}^N \ell_i(\hat{\theta}) - \sum_{i=1}^N \ell_i(\tilde{\theta}) \right],$$

where $\hat{\theta}$ is the unrestricted estimator and $\tilde{\theta}$ is the estimator with Q smooth restrictions imposed.

- Under H_0 ,

$$LR \xrightarrow{d} \chi_Q^2.$$

- The Wald statistic has a disadvantage compared with the score and LR statistics in this (and other nonlinear) contexts: The Wald statistic is not invariant to how the model is parameterized.
- An asymptotic t statistic,

$$t = \frac{(\hat{\theta}_j - a_j)}{se(\hat{\theta}_j)}.$$

is a Wald statistic.

- Suppose we want to test $H_0 : \theta_o = 1$ against $H_1 : \theta_o > 1$. We can use

$$\frac{(\hat{\theta} - 1)}{se(\hat{\theta})}$$

- Can also define $\gamma = \log(\theta)$ and test $H_0 : \gamma_o = 0$. Then $\hat{\gamma} = \log(\hat{\theta})$.
- By the delta method, $\sqrt{N}[\log(\hat{\theta}) - \log(\theta_o)] = \theta_o^{-1}\sqrt{N}(\hat{\theta} - \theta_o) + o_p(1)$, or

$$\sqrt{N}(\hat{\gamma} - \gamma_o) = \theta_o^{-1}\sqrt{N}(\hat{\theta} - \theta_o) + o_p(1).$$

- It follows that

$$Avar(\hat{\gamma}) = Avar(\hat{\theta})/\theta_o^2$$

so

$$se(\hat{\gamma}) = se(\hat{\theta})/\hat{\theta}.$$

- An alternative t statistic is

$$\frac{\hat{\gamma}}{se(\hat{\gamma})} = \frac{\hat{\theta} \log(\hat{\theta})}{se(\hat{\theta})} \neq \frac{(\hat{\theta} - 1)}{se(\hat{\theta})}.$$

- So we have shown that the Wald test is not invariant to nonlinear transformations of the parameters (and corresponding estimators).
- Can show that the score test, if based on the outer product (not a good idea) or the expected Hessian (a better idea) is invariant to reparameterization.
- The LR statistic is invariant because it only uses the maximized values of the objective functions, which will not change by allowable reparameterizations.